SatQuery AI | Earth Observation Evaluation

Earth Observation Mission Division

CLASSIFICATION: OFFICIAL / EO-EVAL

MISSION QUERY & TASK SPECIFICATION

Natural Language Query:	*"What changed between these two dates?"*
Classified Task Type:	Change Detection (Bi-temporal)
Confidence Metric:	**92%** (Dual-Level Verification: Model + Spatial Evidence)
Execution Status:	**COMPLETED (Deterministic DAG Verified)**

EXECUTIVE AI ANALYSIS & REASONING SUMMARY

Primary Finding: Built-up area increased in the eastern section (+1.0% expansion), mainly around the new road corridor and adjacent settlements.

- Built-up Infrastructure: Substantial structural additions (+1.0% / 1.0 km²) concentrated along the eastern transport artery.
- Surface Dynamics: Total detected land-cover transition is 1.5% (1.5 km²) across the 100 km² observation footprint.
- Hydrology & River: The main drainage waterway and riparian embankments maintained strict morphological stability.
- Southern Transition: Peripheral bare land parcels in the south transitioned from fallow ground into active construction sites.

GEOSPATIAL & SENSOR SPECIFICATIONS

Parameter	Specification	Parameter	Specification
Sensor 1 (T1)	Sentinel-2 MSI L2A	Observation Date (T1)	2022-01-15
Sensor 2 (T2)	Sentinel-2 MSI L2A	Observation Date (T2)	2024-06-20
CRS / Projection	EPSG:4326 (WGS 84)	Spatial Resolution	10.0 meters / pixel
Scene Footprint	10 km x 10 km (100 km²)	Center Coordinates	19.0760° N, 72.8777° E

AUDITABLE AGENTIC EXECUTION WORKFLOW

Step	Phase	Engine / Model	Validation / Rationale
01	Query Understanding	Agentic Query Router	Semantic classification to bi-temporal change task
02	Input Validation	Raster & Co-Registration Validator	Verified CRS match, 100% overlap, 10m GSD
03	Model Selection	Dynamic Specialist Registry	Bound RS-ChangeNet & CDVQA specialist tools
04	Change Extraction	RS-ChangeNet (Siamese)	Morphological segmentation, change clusters isolated
05	Evidence Integration	Evidence Fusion Engine	Calculated 92% confidence, 14.2% change area
06	Response Generation	Mission Report Generator	Synthesized visual overlays & PDF/JSON telemetry